

Normal PEPS Fundamental Theorem: Section 3 Route

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1 The source argument

The normal-PEPS part of [MGRPG⁺18] begins after the injective PEPS theorem. Its purpose is not to replace the injective proof, but to make the injective proof applicable after a controlled blocking. A normal PEPS is one for which sufficiently large regions can be blocked into injective tensors. Once such injective regions are available, the proof again applies the three-site injective-chain argument around a chosen edge.

The source proof has three layers. First, Lemma `injective_union` proves that the union of two injective regions is injective. The proof blocks the network into the four regions $A \setminus B$, $A \cap B$, $B \setminus A$, and $(A \cup B)^c$. If a boundary tensor kills the contraction over $A \cup B$, injectivity of A first removes the $A \setminus B$ and $A \cap B$ part, and injectivity of B then removes the remaining $A \cap B$ and $B \setminus A$ part.¹

Second, for the translationally invariant square-lattice theorem, the paper assumes that every 2×3 and 3×2 region is injective. By repeatedly applying the union lemma, the regions denoted R , S , and T in the paper are injective. The regions R and S differ by one site, and their complements are injective when the system is large enough. These are the regions used in the last comparison step.²

Third, the proof of Theorem 3 blocks a neighborhood of each edge into three injective pieces. The red region is A_1 , the blue region is A_2 , and the complement is A_3 . This produces the same three-site injective MPS configuration used in the injective PEPS proof. Lemma `inj_isomorph` therefore gives a gauge on every edge. Translation invariance reduces these

¹The source is `Papers/1804.04964/paper_normal.tex:1324--1400`.

²The regions R , S , and T are drawn at `Papers/1804.04964/paper_normal.tex:1407--1443`.

edge gauges to one horizontal matrix X and one vertical matrix Y . After absorbing them into the second tensor, the final comparison of the R and S regions gives $A \propto \tilde{B}$.³

The general theorem `normal` removes the square-lattice specialization. It assumes precisely the blockings needed above: every edge can be blocked into a three-site injective chain, and every site admits two injective regions with injective complements that differ exactly in that site.⁴

2 Current formal representation

The injective PEPS objects used by the normal theorem are represented in `TNLean/PEPS/Defs.lean`, `TNLean/PEPS/VirtualInsertion.lean`, and `TNLean/PEPS/Blocking.lean`. Chapter 13a of the blueprint now has a separate normal-PEPS section after the injective theorem. This separation is important: the normal theorem needs extra blocking hypotheses before the injective proof route can be invoked.

The present blueprint nodes are the following.

- `lem:peps_injectiveRegion_union` states the union-of-injective regions lemma. It is tracked by GitHub issue #1374.
- `def:peps_normalSquareBlockingRegions` records the square-lattice regions R , S , and T . It is part of GitHub issue #1375.
- `thm:peps_normalEdgeBlockingToInjectiveChain` states that the normal square-lattice hypotheses allow a blocking around every edge into three injective regions. It is part of GitHub issue #1375.
- `thm:peps_tiNormalGaugeAbsorption` records the translationally invariant gauge absorption with horizontal gauge X and vertical gauge Y . It is part of GitHub issue #1375.
- `thm:fundamentalTheorem_normalSquarePEPS` records the square-lattice theorem corresponding to Theorem 3. It is part of GitHub issue #1375.
- `thm:fundamentalTheorem_normalPEPS` records the general normal PEPS theorem following Theorem 3. It is part of GitHub issue #1375.

³The proof of Theorem 3 is `Papers/1804.04964/paper_normal.tex:1449--1572`.

⁴The general theorem is `Papers/1804.04964/paper_normal.tex:1574--1585`.

The normal route itself is tracked by GitHub issue #1373. The normal-section diagram work is tracked by GitHub issue #1376. The general PEPS status tracker is GitHub issue #1252.

3 Remaining mathematical obligations

The normal PEPS theorem cannot be obtained by citing the injective theorem alone. The following additional obligations remain.

1. Define a formal notion of injectivity for a blocked region, not only for a single vertex. The statement must be able to express the regions $A \setminus B$, $A \cap B$, $B \setminus A$, R , S , T , and complements of such regions.
2. Prove the union-of-injective-regions lemma. The proof must follow the source proof by applying the left inverses for A and B in sequence.
3. Package the square-lattice normal hypotheses: every 2×3 and 3×2 region is injective, and the ambient size is large enough for the required complements.
4. Prove that the paper's regions R , S , and T are injective under those hypotheses. This is where the union lemma is used repeatedly.
5. Prove that the edge-centered red, blue, and complementary regions form a three-site injective chain around every edge. This is the normal analogue of the injective edge-blocking bridge tracked by GitHub issue #1366.
6. Reuse the injective-chain insertion route after this blocking: GitHub issue #1367, GitHub issue #1368, GitHub issue #1363, GitHub issue #1364, GitHub issue #1361, and GitHub issue #1360.
7. In the translationally invariant case, show that the edge gauges obtained after blocking reduce to one horizontal gauge X and one vertical gauge Y , up to scalar.
8. Prove the scalar condition $\lambda^{nm} = 1$ for the square-lattice theorem, and state separately the scalar freedom in the non-translation invariant normal theorem.

4 Source-to-formalization ledger

- Lemma `injective_union`: blueprint `lem:peps_injectiveRegion_union`, issue #1374.
- Regions R , S , and T : blueprint `def:peps_normalSquareBlockingRegions`, issue #1375.
- Edge blocking into red, blue, and complementary injective regions: blueprint `thm:peps_normalEdgeBlockingToInjectiveChain`, issue #1375.
- Application of Lemma `inj_isomorph` after normal blocking: downstream issues #1367, #1368, #1363, and #1364.
- Translationally invariant gauge absorption: blueprint `thm:peps_tinormalGaugeAbsorption`, issue #1375.
- Final R and S comparison: blueprint `thm:peps_oneVertexComplementComparison`, issue #1360.
- Theorem 3: blueprint `thm:fundamentalTheorem_normalSquarePEPS`, issue #1375.
- General theorem `normal`: blueprint `thm:fundamentalTheorem_normalPEPS`, issue #1375.

5 Blueprint diagrams

The normal route now has its own semantic tensor-network commands. These are not substitutes for the injective theorem diagrams; they attach the additional normal blocking layer to named proof nodes.

- `\TNPEPSInjectiveRegionUnion` depicts the four regions used in Lemma `injective_union`.
- `\TNPEPSNormalRegionsRS` depicts the regions R and S from the square-lattice proof.
- `\TNPEPSNormalRegionT` depicts the region T .
- `\TNPEPSNormalEdgeBlockingReduction` depicts the red/blue/complementary blocking around a distinguished edge and the resulting three-site chain.

- `\TNPEPSTINormalGaugeAbsorption` depicts the final translationally invariant local gauge formula with horizontal gauge X and vertical gauge Y .

All of these commands are registered in `blueprint/src/Packages/tn_diagrams.py`, so the PDF and web blueprint use the same public diagram names.

6 Conclusion

The normal PEPS route is now separated from the injective PEPS route in the blueprint and in the GitHub issue tree. The formal proof still remains open: the next genuine mathematical step is to introduce blocked-region injectivity and prove Lemma `injective_union`. Once that lemma and the normal edge-blocking hypotheses are formalized, the downstream proof should reuse the injective edge-insertion route rather than duplicate it.

References

- [MGRPG⁺18] András Molnár, José Garre-Rubio, David Pérez-García, Norbert Schuch, and J. Ignacio Cirac. Normal projected entangled pair states generating the same state. *New Journal of Physics*, 20(11):113017, 2018.